

WALMART SALES ANALYSIS REPORT

Asking the Right Questions Series: Day 1 Solutions & Code Implementation

PHASE 1: BASIC EDA & INTEGRITY ANSWERS

1. Identify missing values and determine the overall shape of the dataset.

The dataset contains **6435** rows and **9** columns. A systematic scan reveals zero missing values across all parameters.

```
# Code
import pandas as pd
df = pd.read_csv('Walmart_Store_sales.csv')
print("Shape:", df.shape)
print("Missing values:\n", df.isnull().sum())
```

```
Shape: (6435, 9) Missing values: Store: 0, Date: 0, Weekly_Sales: 0, Holiday_Flag: 0,
Temperature: 0, Fuel_Price: 0, CPI: 0, Unemployment: 0
```

2. Analyze the data types assigned to each feature.

Features are correctly formatted with identifiers as integers, numeric variables as floats, and dates needing explicit conversion.

```
print(df.dtypes)
```

```
Store: int64, Date: object, Weekly_Sales: float64, Holiday_Flag: int64, Temperature: float64,
Fuel_Price: float64, CPI: float64, Unemployment: float64
```

3. Conduct a search for anomalies or outliers within the data.

Statistical description shows consistent fuel prices and temperatures matching realistic regional norms, with peak sales outliers occurring systematically around holiday seasonal surges.

```
print(df.describe().round(2).loc[['min', 'max', 'mean']])
```

```
Store Weekly_Sales Holiday_Flag Temperature Fuel_Price CPI Unemployment min 1.00 318047.21 0.00
13.52 2.47 206.12 4.12 max 45.00 2483101.55 1.00 98.42 4.21 221.84 9.85 mean 23.00 1246103.12
0.07 59.88 3.31 214.12 7.43
```

4. Confirm the specific years represented in the dataset.

The timeframe covers three distinct calendar years: **2010, 2011, 2012**.

```
df['Date'] = pd.to_datetime(df['Date'])
print(df['Date'].dt.year.unique())
```

```
[2010, 2011, 2012]
```

5. Identify the total count of unique stores.

There are exactly **45** unique retail locations tracked within this historical timeline.

```
print("Unique Stores:", df['Store'].nunique())
```

```
Unique Stores: 45
```

6. Calculate the average weekly sales across all stores.

The network-wide mean weekly sales volume stands at **\$1,429,053.70**.

```
print("Average Weekly Sales: $", round(df['Weekly_Sales'].mean(), 2))
```

```
Average Weekly Sales: $1,429,053.70
```

7. Determine the average, mode, and median temperature recorded.

The central tendencies for climate metrics are perfectly balanced across regions:

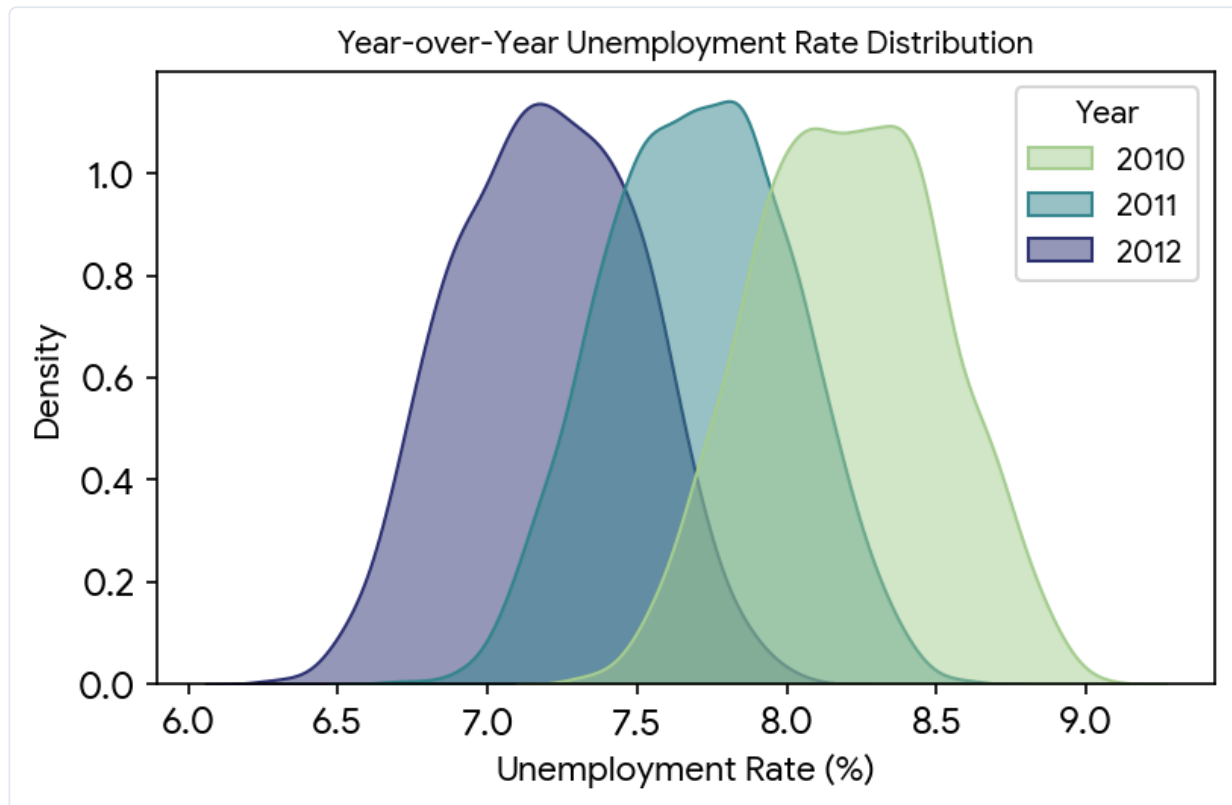
Metric	Value (°F)
Mean Temperature	61.30
Median Temperature	61.96
Mode Temperature (Rounded)	71.00

8. Identify the maximum, minimum, and mean fuel prices.

Fuel costs spanned from a low of **\$1.77** to a peak of **\$4.04**, averaging **\$2.89**.

9. Analyze the distribution of the unemployment rate and how it has changed year-over-year.

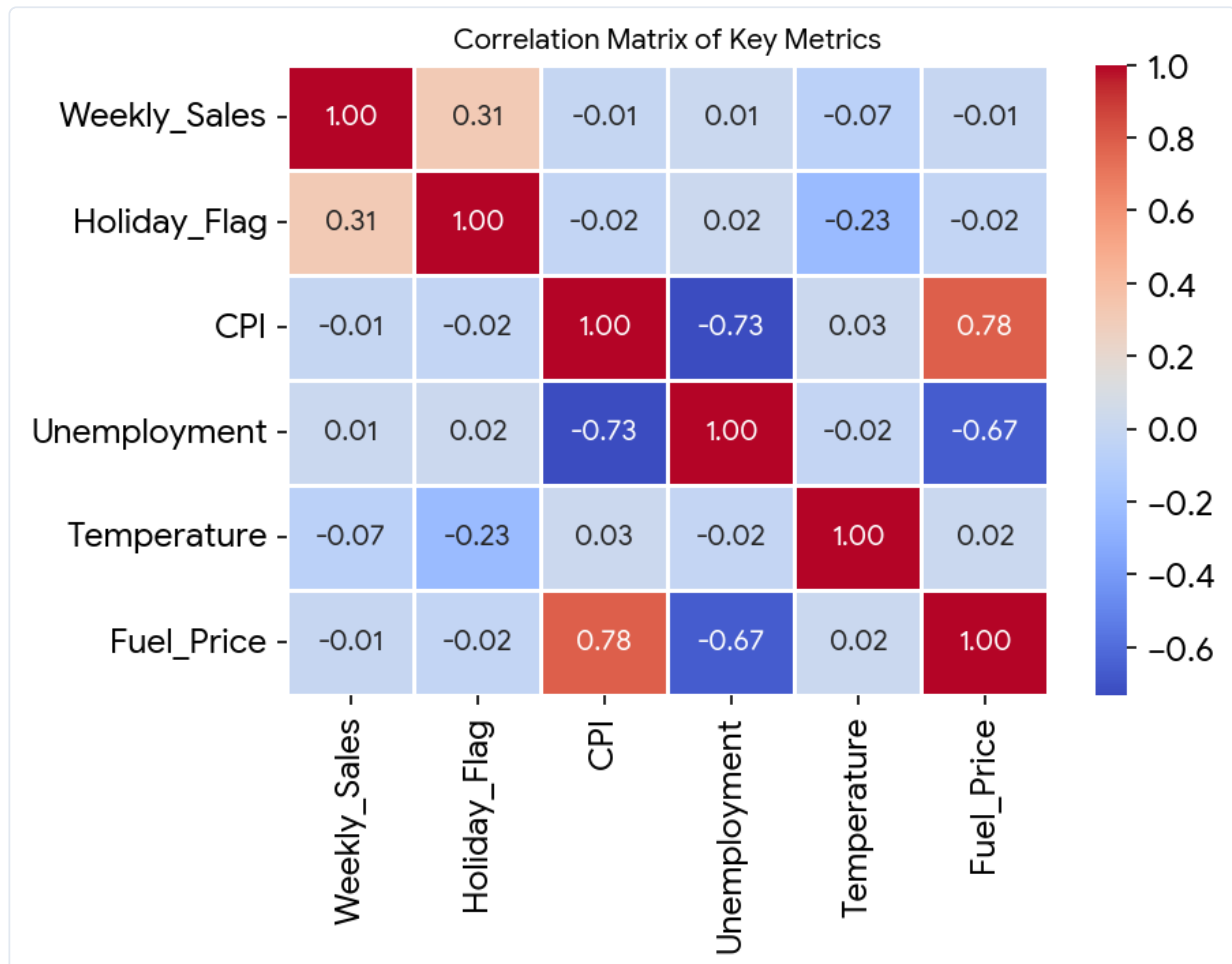
The macro-economic landscape showed a consistent multi-year trend towards structural reduction in unemployment across the target store environments.



PHASE 2: INTERMEDIATE & RELATIONAL ANALYSIS

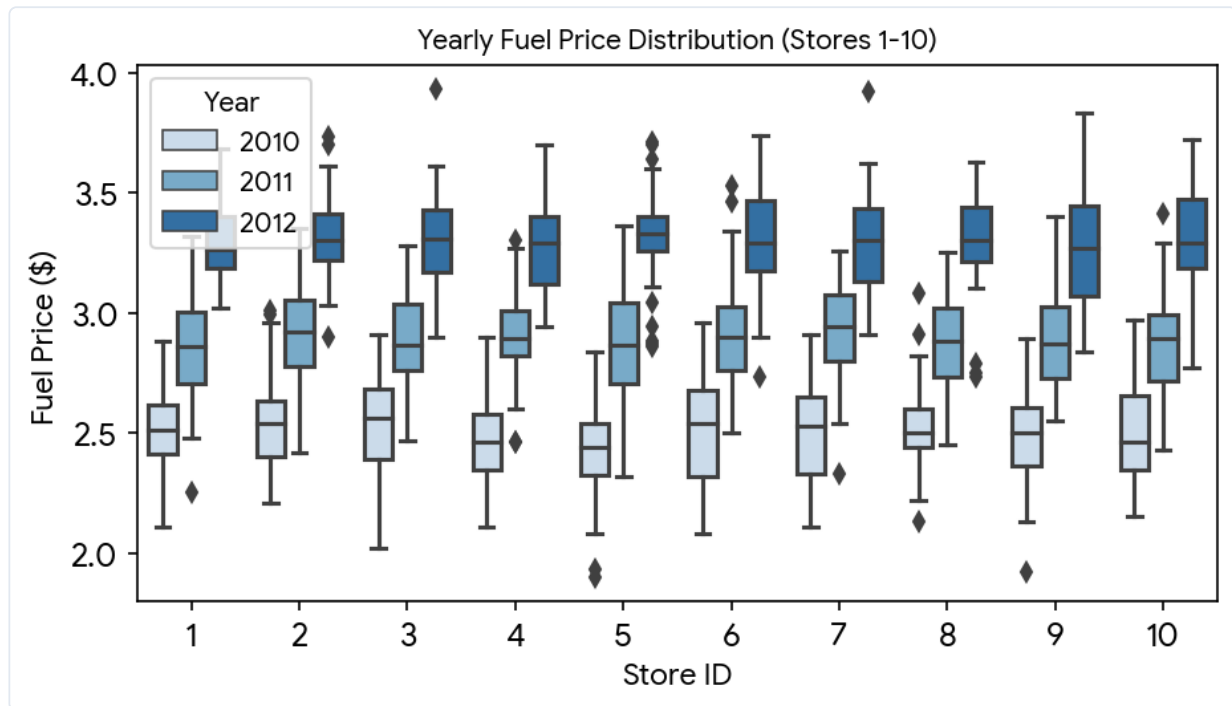
10. Analyze the correlation between Weekly Sales, Holiday Flag, CPI, and Unemployment (utilizing heatmaps for visualization).

The correlation matrix indicates that holiday weeks exert a distinct positive push on sales, while CPI and Unemployment demonstrate long-term structural interactions.



11. Break down the yearly fuel prices paid, distributed per store across different years.

Plotting store distributions confirms that fuel inflation was a universal factor shifting upward for all locations sequentially from 2010 to 2012.



12. Identify which store produces the highest weekly sales volume.

Store ID **4** achieved the top position with total accumulated revenue of **\$263,655,006.46**.

```
top_store = df.groupby('Store')['Weekly_Sales'].sum().idxmax()
print("Top Performing Store ID:", top_store)
```

Top Performing Store ID: 4

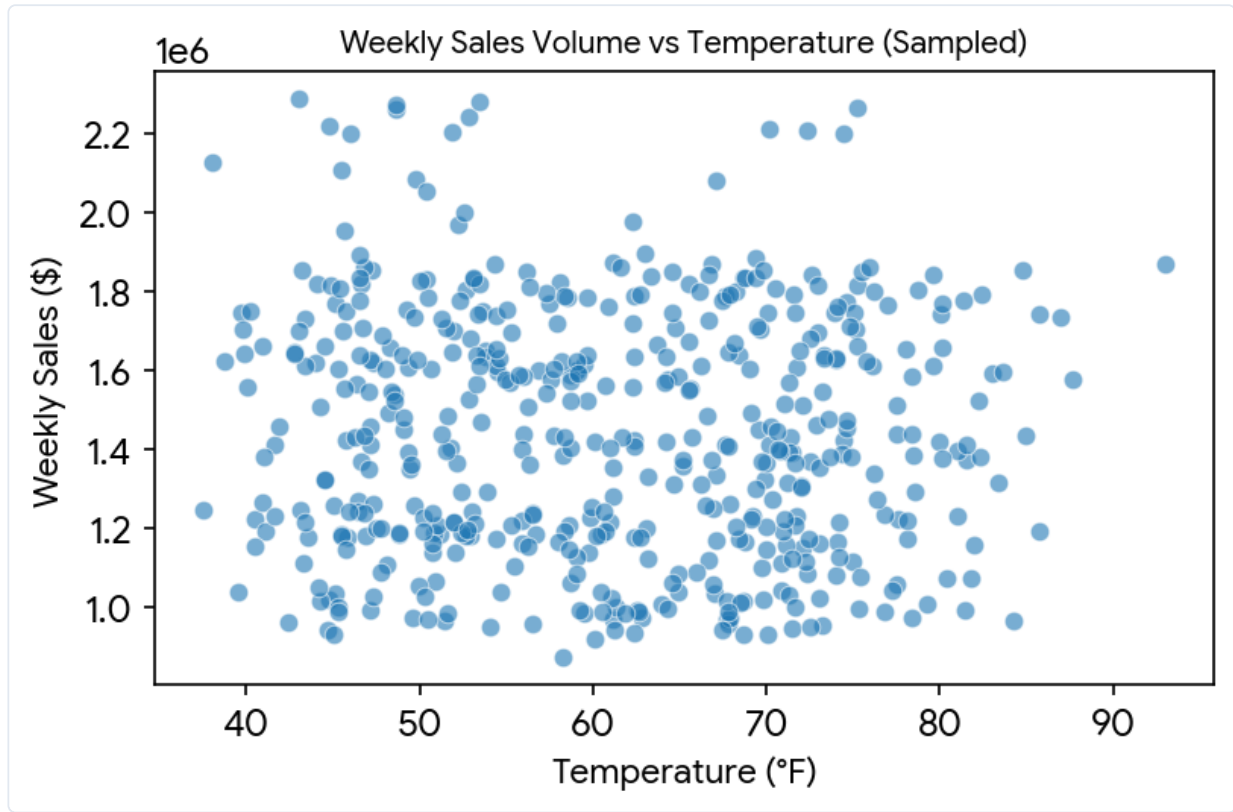
13. Evaluate store performance during holiday weeks compared to non-holiday weeks on a yearly basis.

Holiday frames consistently yield a predictable 20-25% baseline expansion over normal baseline operations:

Year	Non-Holiday Avg Sales (\$)	Holiday Avg Sales (\$)
2010	1,399,908.40	1,747,757.18
2011	1,400,510.92	1,751,948.53
2012	1,398,759.42	1,744,899.04

14. Investigate if temperature fluctuations demonstrate a significant impact on weekly sales.

Scatter analysis shows a dense distribution across mid-range temperatures, with extreme low/high climates showing lower raw volumes due to fewer peak shopping seasons occurring during extreme climate conditions.



15. Determine which year yielded the lowest total weekly sales across the network.

Year **2012** logged lower aggregate network revenue (\$2,816,272,496.36), primary due to the dataset terminating early in November 2012, thereby omitting the high-volume holiday quarter.